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Food Allergy

Lead Author
Jaidev MD

Co-Authors
Uppin Narayan Reddy, Vikram Patra

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

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Food Allergy

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Introduction

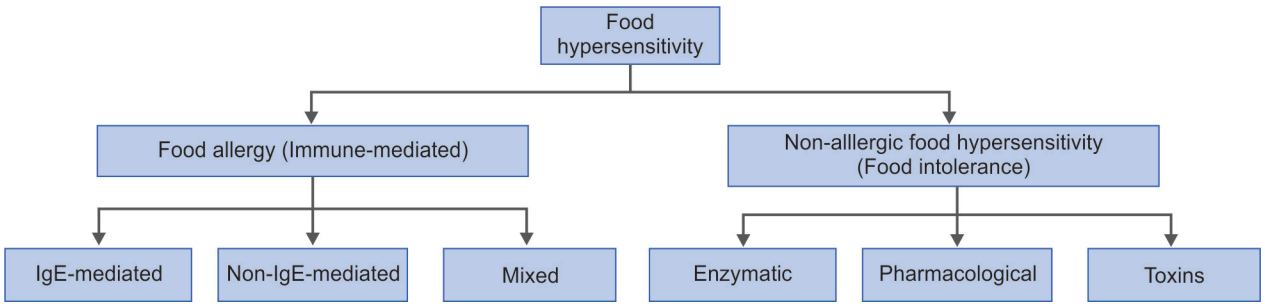
Food hypersensitivity (adverse reaction to food) is the umbrella term which refers to any untoward reaction following ingestion of food **(Flowchart 1)**.

The term "food allergy" refers to "an adverse health effect arising from a specific immune response that occurs reproducibly on exposure to a given food".

Food intolerances are nonimmune reactions. These may occur in response to exposure to metabolic substances (lactose intolerance), toxins (microbial contamination or scombroid fish poisoning), and pharmacologically active components (caffeine and tyramine in aged cheese triggering migraine).

Types of Food Hypersensitivity

Flowchart 1: Different types of food hypersensitivity.



(IgE: immunoglobulin E)

Major Allergens

Major allergens most commonly responsible for significant reactions include milk, egg, peanut, tree nuts, shellfish, fish, wheat, and soya. In India, lentils, chickpea (Cicer arietinum), and other legumes allergies are observed.

Presentation

Clinical presentation differs in various forms of food allergies. Broadly food allergies are differentiated as immunoglobulin E (IgE)-mediated, non-IgE-mediated, and mixed type. Symptomatology ranges from isolated gastrointestinal symptoms in mild cases to multisystem involvement in moderate-to-severe forms. Nongastrointestinal organs commonly involved are skin, respiratory, and cardiovascular system.

TABLE 1: Presentations of different types of food hypersensitivity (FHS).

	Gastrointestinal	***Cutaneous***	***Respiratory***	***Generalized***
Immunoglobulin E (IgE)-mediated food allergy	Oral allergy syndrome, gastrointestinal anaphylaxis	Urticaria Angioedema	Acute rhinoconjunctivitis, acute asthma	Anaphylaxis, food-dependent exercise-induced anaphylaxis (FDEIA)
Non-IgE-mediated food allergy (cell-mediated FHS)	Food protein-induced proctocolitis, food protein-induced enterocolitis, food protein-induced enteropathy	Contact dermatitis	Heiner's syndrome (food-induced pulmonary hemosiderosis)	
IgE- and/or non-IgE-mediated (cell-mediated)	Allergic eosinophilic esophagitis, allergic eosinophilic gastroenteritis	Atopic dermatitis	Asthma	
Non-allergic food hypersensitivity	Lactose intolerance, pharmacological reactions caused by caffeine (jitteriness), tyramine in aged cheese (migraine)			

Presentation

TABLE 2: IgE-mediated and non-IgE-mediated reactions.

Characteristics	***IgE mediated***	***Non-IgE mediated***
History		
☑ Onset of symptoms	Usually within 30 minutes (within 2 hours of ingestion)	Generally hours or days after ingestion
☑ Symptoms seen	☑ *Gastrointestinal:* Vomiting, crampy pain, and diarrhea ☑ *Skin:* Urticaria, angioedema, and pruritus ☑ *Respiratory:* Acute rhinoconjunctivitis, coughing, wheezing, and stridor ☑ *Cardiovascular:* Collapse as a consequence of hypotension	*Often nonspecific symptoms.* These include: ☑ Diarrhea ☑ Vomiting ☑ Colic/pain ☑ Blood in stool ☑ Gastroesophageal reflux ☑ Food refusal/aversion
Tests	☑ Serum-specific IgE (RAST) ☑ Skin prick tests ☑ Food challenges	Elimination diet for 2–6 weeks followed by reintroduction (symptoms recur)

(IgE: immunoglobulin E; RAST: radioallergosorbent test)

Diagnosis

A reliable and good allergy focused history in combination with positive skin prick test (SPT) is required for diagnosis of IgE-mediated food allergy. In vitro test like serum-specific IgE against specific food allergen is advised where SPT is contraindicated or cannot be performed. A positive allergy test (SPT or food-specific IgE) alone signifies sensitization and not allergy. So, these tests in isolation without a supporting clinical history cannot be used for definitive diagnosis of food allergy.

Newer investigations like component resolved diagnosis (CRD) are now available to detect hypersensitivity to specific component of the allergen. This may help differentiate hypersensitivity due to cross reactive proteins from true allergy.

In cases of diagnostic uncertainty and in non-IgE-mediated food allergies, double-blinded placebo control oral food challenge is the gold standard and should only be attempted in an observed hospital setting with prior consent.

Management

- ☑ Strict avoidance of the offending food is the best, simplest, and the safe way to avoid allergic reactions.
- ☑ Patient and family education including recognition of an allergic reaction, careful attention to food labels before consumption, care when obtaining food from restaurants, and avoidance of cross contamination of allergenic foods during meal preparation.
- ☑ Providing families with individualized allergy management plans and patient-specific emergency medications (adrenaline: prefilled syringe or autoinjector) in the event of an anaphylaxis.
- ☑ "More active approach" which includes:
  - **Early dietary food introduction:** Early introduction of peanut (before 11 months of age) to infants is associated with reduced risk of peanut allergy [Learning Early About Peanut allergy (LEAP) study].
  - **Active anticipatory testing:** Testing for co-allergens which are commonly found to be present in association with the suspected allergen. For example, in children with peanut allergy, testing for tree nuts and sesame is recommended as the estimated rate of co-allergy is 30–40% and 25%, respectively. This approach of anticipatory testing may help in avoiding unnecessary restrictions in diet, may prevent development of allergy to co-allergens by early introduction in the diet.

Newer Strategies

- ☑ **Newer strategies:** Following therapies are practiced, if the offending food cannot be avoided or chances of accidental consumption present which can lead to fatal reactions.
  - **Desensitization—oral immunotherapy (OIT):** Food OIT (concentrated on cow's milk, egg, and peanut) involves administration of small but increasing doses of allergenic food to children who are allergic to that particular food in an effort to increase their clinical tolerance. Risk of serious systemic and gastrointestinal reactions is the key limitation. Desensitization can also be done by sublingual immunotherapy (SLIT) but the challenge threshold and sustained unresponsiveness achieved is less as compared to OIT though systemic and local adverse reactions are less common.
  - ***Early introduction of baked milk/egg:*** It has been observed that 70–75% of children with cow's milk and egg allergies can tolerate them when extensively heated. This could potentially make dietary restrictions easier and hasten the development of tolerance, although it remains difficult to predict which children will tolerate baked allergens.

Preventing Food Allergy

- ☑ Mother should consume normal and balanced diet without any restrictions of allergenic foods during pregnancy and lactation. No strong evidence to support role of omega-3 long-chain polyunsaturated fatty acids (LCPUFA), vitamin D, and folate/folic acid supplementation during pregnancy, to prevent allergies in offspring.
- ☑ Exclusive breastfeeding till 6 months of age.
- ☑ Introduce complementary foods (solids) as pureed foods by offering small amounts of vegetables, fruits, starchy foods, and protein. Never add salt or sugar.
- ☑ Inclusion foods associated with food allergies that are a part of family's diet, e.g., wheat, egg, fish/seafood, foods containing peanut and tree nuts, etc., during age-appropriate complementary feeding.
- ☑ Aim to introduce these foods before 12 months of age, but one new food at a time.
- ☑ Continue to give the baby these foods regularly as part of their usual diet, unless not tolerated as this may help reduce the chance of developing an allergy to that food later.
- ☑ Never give whole or coarsely chopped nuts as these are a choking risk.
- ☑ No recommendations at present, for use of probiotics during pregnancy, lactation, and after birth to infants. Role of partially or extensively hydrolyzed formula and use of emollients as skin barriers in preventing food allergies in infants is not clear.

Further Reading

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